Formulas for energy and motion

3a

Speed up your calculations by using a calculator.

1 There are a number of formulas that are used when calculating things like speed, distance and acceleration. Let's see how good you are with some of those formulas. Connect the matching items (quantity, symbol, formula, unit, comment) in the table below by drawing lines between them.

Quantity	Symbol	Formula	Unit	Comment
time	a	d/t	J	acceleration due to earth's gravity
speed	d	d/v	J	amount of matter
potential energy	F	F/a	kg	change in speed per unit time
mass	g	ma	m	energy due to height
kinetic energy	h	mgh	m	energy due to movement
height	KE	$^1/_2\,mv^2$	m/s	height above sea level
gravity acceleration	m	vt	m/s^2	how far
force	PE	$(v_2 - v_1)/t$	m/s^2	how fast
distance	v	PE/(mg)	N	how long
acceleration	t	PE/(mh)	s	strength

Note: potential energy = gravitational energy and kinetic energy = movement energy

2 Use formulas from question 1 to answer the following questions. Write the correct formula, show your working and give your answer as a number with units.

a It takes a sports car 7 s to reach a speed of 100 km/h (= 28 m/s). What is the car's acceleration?

b The sports car has a mass of one tonne. How much kinetic energy would it possess at 100 km/h (= 28 m/s), at 50 km/h (= 14 m/s), at 25 km/h (= 7 m/s) and at 0 km/h?

c The sports car speeds along at 100 km/h for 5 s. What distance does it cover? Use the m/s speed.

d The car travels another 200 m at that speed. How long does it take to cover the 200 m?

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e If it continued straight ahead, the car would leave the road and plummet 200 m downwards. Assume that the acceleration due to gravity is 10 m/s^2. Find the car's potential energy relative to the bottom of the 200 m fall.

f The car, travelling at 100 km/h, brakes hard and comes to a stop in 4 s. Assuming constant deceleration, what is the deceleration during the stopping time? Use the m/s speed.

g Another similar car, travelling at the same speed, does not fare as well. It hits a wall at 100 km/h and comes to a fairly sudden stop in half a second (fortunately the car has crumple zones which stop the car more slowly). What is its acceleration? Use m and s.

3 A cyclist of mass 50 kg rides a pushbike of mass 15 kg. The cyclist is riding at a speed of 72 km/h (= 20 m/s).

a What is the kinetic energy of the cyclist and bike?

b What distance does the cyclist cover in 30 s?

c At a constant speed of 72 km/h how long would it take the cyclist to ride 1 km?

d The cyclist brakes and stops in 4 s. What is the cyclist's deceleration?

e Assume that the force of the cycle tyres on the road is uniform during the stopping period. What force is acting to stop the cyclist?

f Imagine that this cyclist and bike were to fall off a building. How high would the building have to be in order to provide the bike and cyclist with the same amount of energy that they had when travelling at 72 km/h?

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